Coaxial Two Way Switch (DPDT) | BN 512690C0001

Typical illustration

Radio frequency characteristics

Interface type (4 connections)	7-16 female (50 Ω) according to IEC 61169-4					
Characteristic impedance	50 Ω					
Frequency range	0 to 1 MHz	1 to 3 MHz	3 to 10 MHz	10 to 30 MHz	30 to 60 MHz	60 to 100 MHz
VSWR, max.	1.04	1.04	1.04	1.04	1.04	1.04
Isolation, min.	80 dB	80 dB	80 dB	80 dB	80 dB	80 dB
Insertion loss, max.	0.05 dB	0.05 dB	0.05 dB	0.05 dB	0.05 dB	0.05 dB
Average power capability RFC1) at ambient temperature -10 to +45°C	36 kW	22.4 kW	14.2 kW	9.9 kW	8.0 kW	6.4 kW
Frequency range	0.1 to 1 GHz	1 to 2 GHz	2 to 3 GHz	3 to 4 GHz	4 to 5 GHz	5 to 6 GHz
VSWR, max.	1.04	1.08	1.08	1.12	1.20	1.35
Isolation, min.	80 dB	80 dB	80 dB	60 dB	50 dB	40 dB
Insertion loss, max.	0.05 dB	0.05 dB	0.10 dB	0.10 dB	0.10 dB	0.20 dB
Average power capability RFC1) at ambient temperature -10 to +45°C	2.0 kW	1.4 kW	1.1 kW	1.0 kW	0.9 kW	0.8 kW
Peak voltage capability RFC1)	4.0 kV					

RFC1) Standard conditions: - Dielectric: Dry air under standard pressure at sea level (p = 1013 hPa)
- Load VSWR, max. 1.0 (no standing wave)
- No modulation, sinusoidal carrier only

Electrical and mechanical characteristics

Switch type		Two way switch, DPDT
Actuator type		Solenoid drive, latching, self cutoff
Connector 1 EMC1) for operating voltage, control, interlock contacts and signaling		25 pole connector according to DIN 41652 / IEC 807-2
Operating	Operating voltage	12 V DC ±10%
	Operating current, typ. EMC2)	2 A
	Standby current, max. EMC2)	25 mA
	Nominal fuse	The switch must be secured externally by a time-delay fuse, 2 A
Control	Control voltage	U In LOW = 0 to 4 V DC / -0.7 mA (0 - active) U In HIGH = 8 to 32 V DC
	Current limiting	The circuit must be limited externally to 0.5 A
	Lead time typ. EMC2)	5 ms (the interlock contacts open 5 ms before and

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Signal contacts Interlock contacts	(only interlock contacts)	close 2 ms after switching of the RF contacts)
	Maximum ratings	ES1 circuits according to EN 62368-1, 42.4 V ACpk / 60 V DC / 0.5 A
	Current limiting	The circuit must be limited externally to 0.5 A
Switching time, typ. EMC2)		100 ms
Command hold time, min.		100 ms (during this time, the voltage at control input must not change)
Switching frequency, max.		30 operations per minute
Lifetime, min.		500,000 operations
Weight, approx.		1.2 kg

EMC1) Suitable mating connector included
EMC2) At room temperature and nominal voltage 12 V DC

Environmental conditions

Operational conditions	ETSI EN 300 019-1-3 V2.3.2 (2009-1) class 3.1 N
Ambient temperature EC1)	-10 to +60°C
Condensation	Not allowed
Relative humidity, max.	95%

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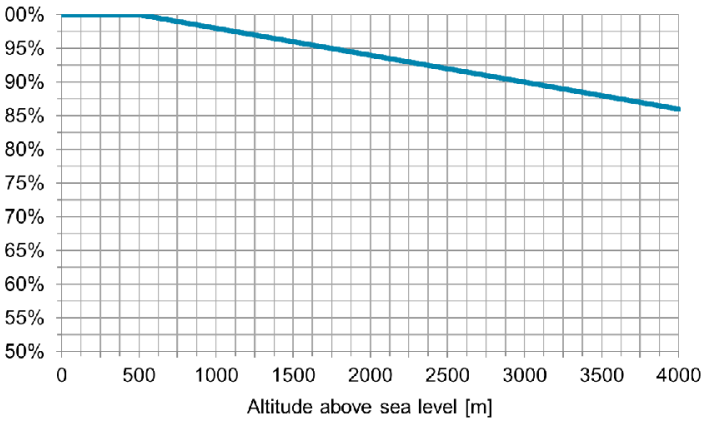
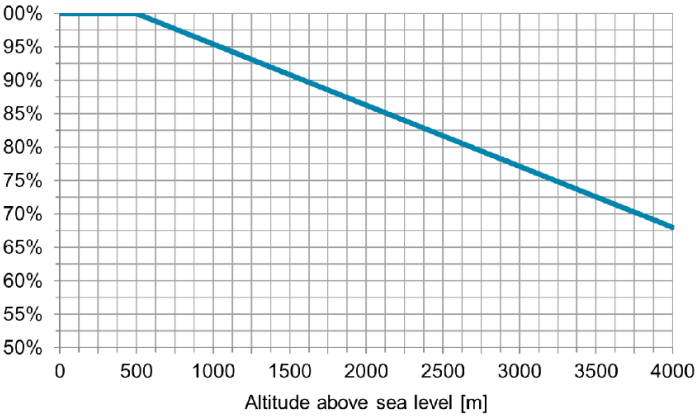
Derating of input power with increasing altitude	The maximum input power can be applied up to 500 m or 1600 ft above sea level unless noted otherwise in the data sheet. Above this height the maximum input power must be reduced as shown in the diagram. 
Derating of voltage with increasing altitude	The maximum voltage can be applied up to 500 m or 1600 ft above sea level unless noted otherwise in the data sheet. Above this height the voltage must be reduced as shown in the diagram. 

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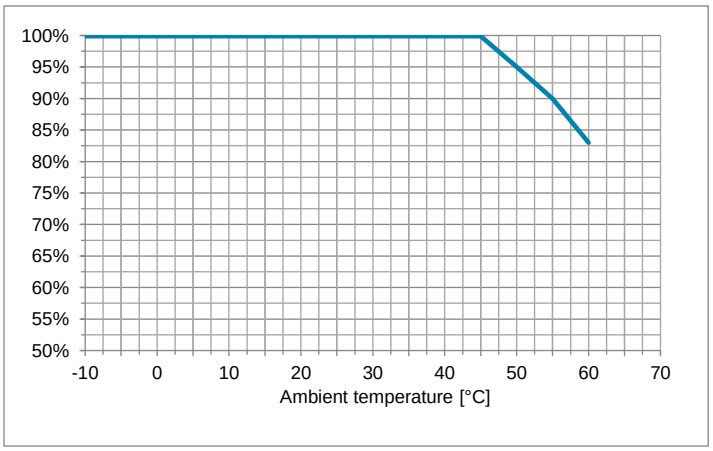
Derating of input power with increasing ambient temperature	The maximum input power can be applied up to +45°C ambient temperature unless noted otherwise in the data sheet. Above this ambient temperature the maximum input power must be reduced as shown in the diagram. 
Max. altitude above sea level	4,000 m or 13,120 ft according to IEC EN 60664-1
Protection class	III according to IEC EN 61140
IP protection level	IP40 according to IEC EN 60529 (all interfaces terminated)
Installation position	Any
Transport conditions	ETSI EN 300 019-1-2 V2.1.4 (2003-04) class 2.2
Ambient temperature	-25 to +70°C
Rain, condensation, icing	Not allowed
Storage conditions	ETSI EN 300 019-1-1 V2.1.4 (2003-04) class 1.2
Ambient temperature	-10 to +60°C
Rain, condensation, icing	Not allowed

EC1) Extended temperature range on request

Applicable documents

Product manual	M36311

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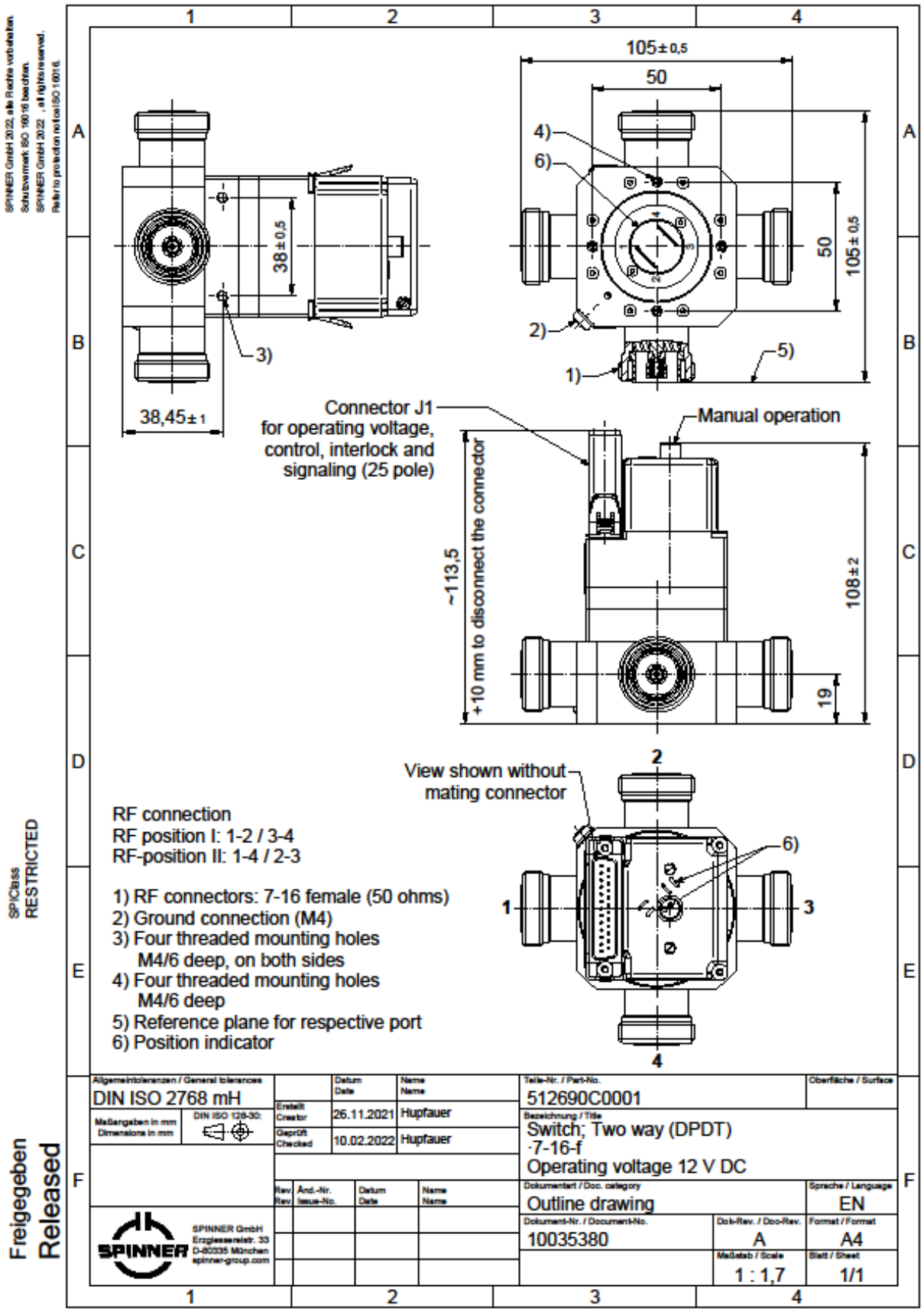
Outline (all dimensions in millimeter)



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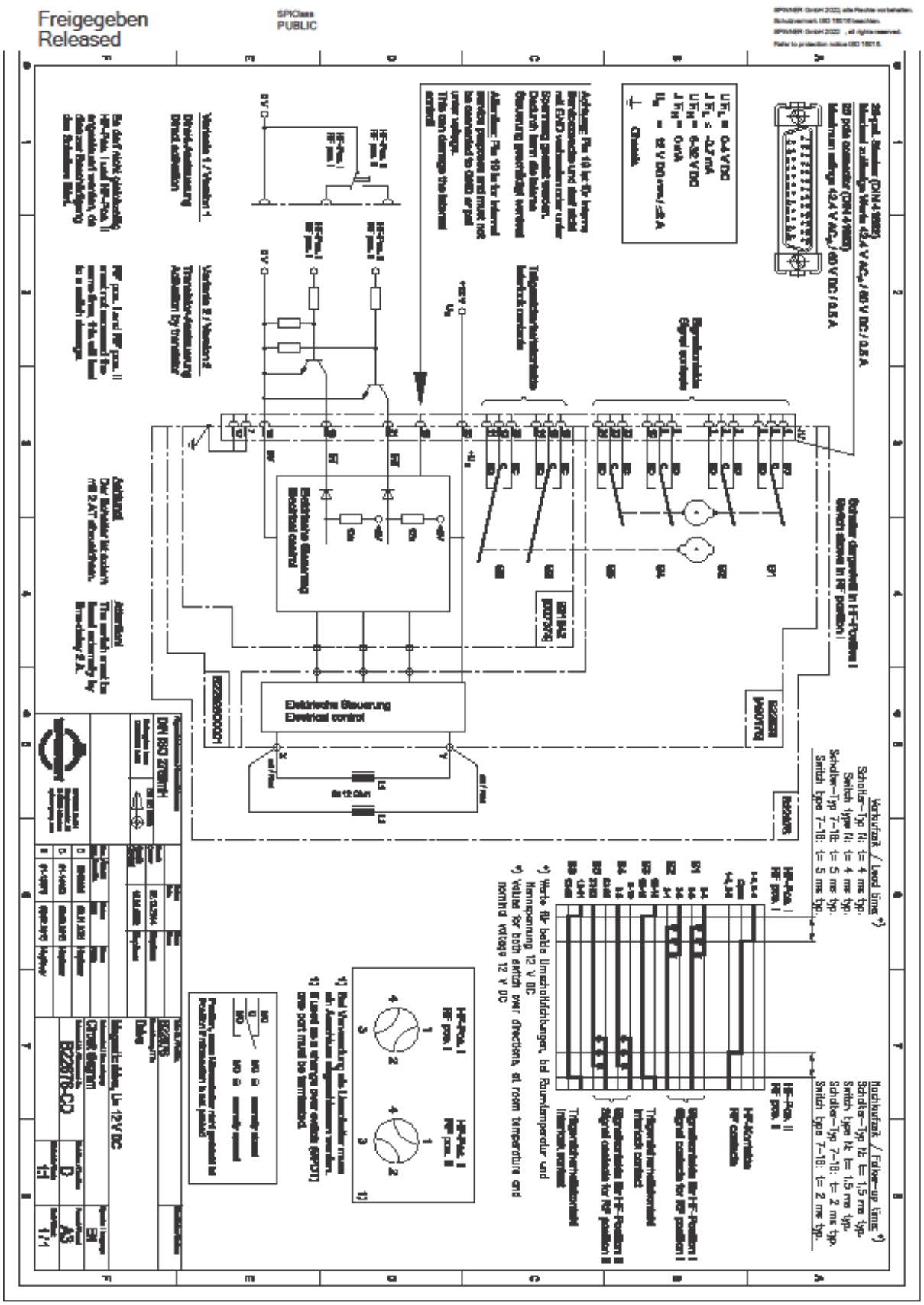
Circuit diagram



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